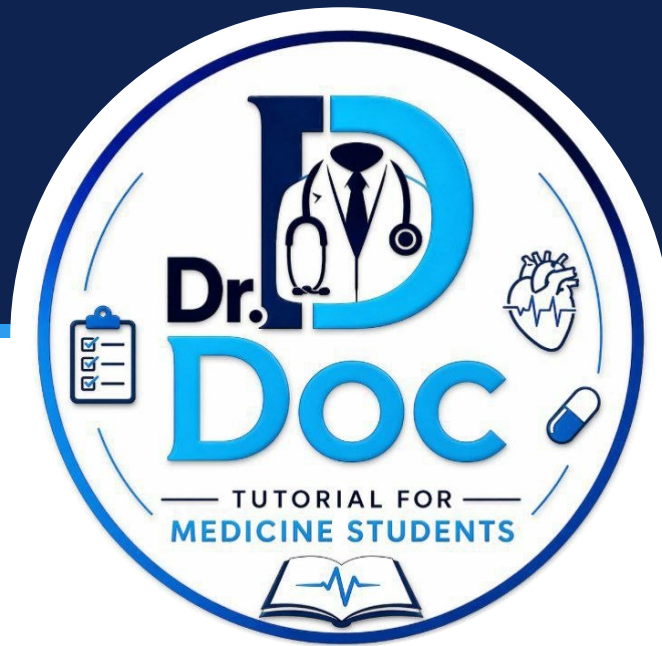




Introduction to Medicine Mock Exam — Answer Guide

122 questions across histology, anatomy, physiology,
embryology and biochemistry, with the key point behind every answer



18 topic areas, first-round mock exam:

Histology	Osteology & Skeletal Anatomy	Neuroanatomy
General & Regional Anatomy	Cardiovascular Anatomy	Cell Membrane & Transport
Endocrine & Renal Physiology	Reproductive Physiology	Gastrointestinal Physiology
Autonomic & Nervous System Physiology	Muscle Physiology	Respiratory Physiology
Reproductive Anatomy	Embryology	Biochemistry: Carbohydrates
Biochemistry: Membranes & Cell Biology	Biochemistry: Enzymes	Biochemistry: Proteins & Lipids

122

questions

18

topic areas

A-F

choices covered

Dr. Doc Tutorials
Tutorial for Medicine Students

Introduction to Medicine - Mock Exam

122 questions, first round, with the correct answer and a key point for every question

Prepared by Dr. Doc Tutorial

Quick answer key

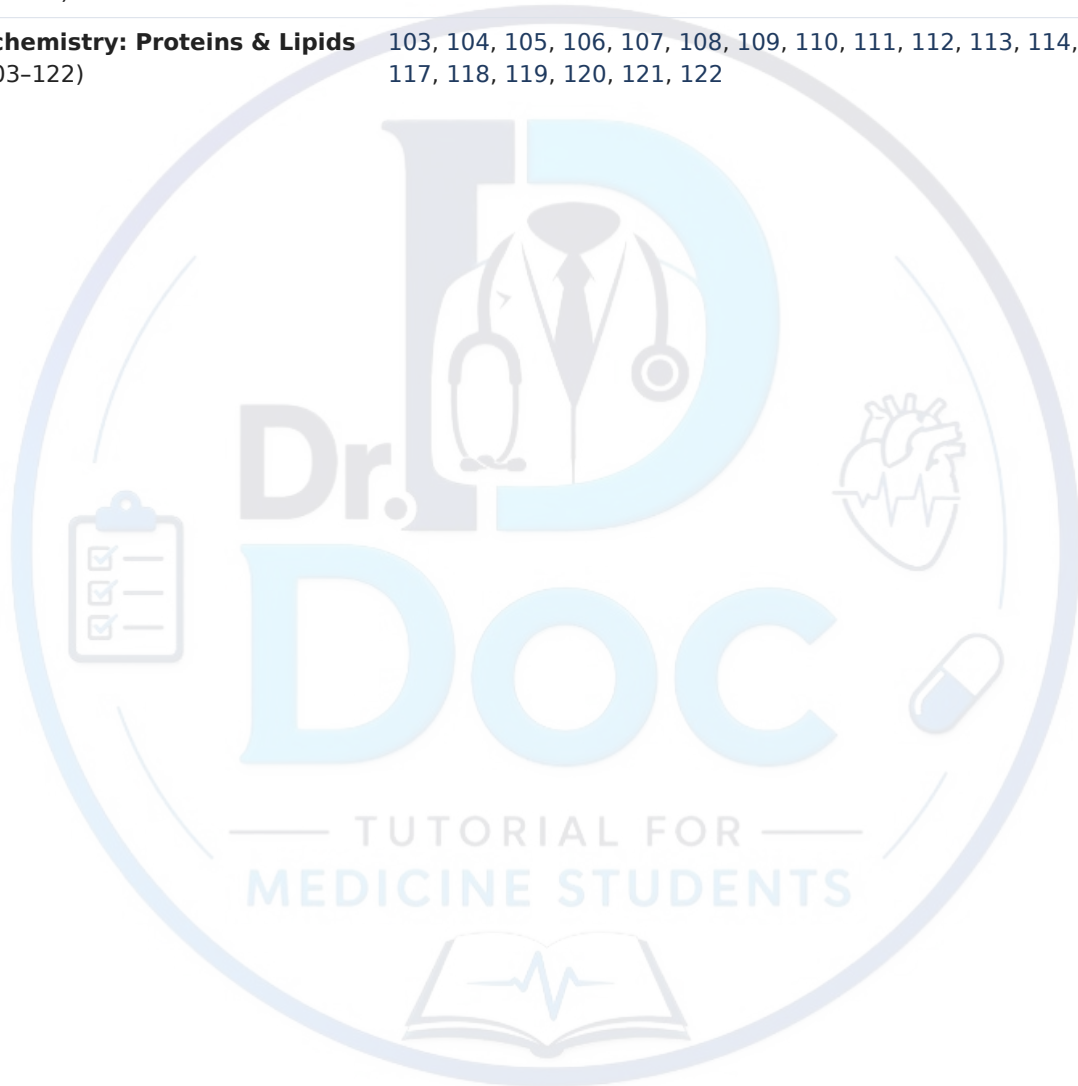
Tap a number to jump to its explanation. Colors show the topic area.

1 C	2 B	3 A	4 C	5 A	6 D	7 E	8 C	9 C	10 A
11 C	12 B	13 B	14 C	15 B	16 B	17 B	18 C	19 B	20 C
21 E	22 D	23 E	24 C	25 A	26 D	27 C	28 B	29 C	30 B
31 C	32 D	33 B	34 C	35 C	36 B	37 C	38 A	39 B	40 B
41 B	42 B	43 B	44 B	45 B	46 B	47 C	48 B	49 B	50 B
51 A	52 A	53 A	54 E	55 A	56 E	57 C	58 C	59 E	60 B
61 C	62 C	63 B	64 D	65 A	66 E	67 B	68 D	69 B	70 C
71 B	72 B	73 B	74 B	75 B	76 D	77 B	78 C	79 B	80 C
81 B	82 C	83 B	84 B	85 C	86 C	87 B	88 B	89 B	90 B
91 A	92 C	93 B	94 A	95 B	96 B	97 B	98 B	99 B	100 B
101 B	102 B	103 B	104 A	105 B	106 B	107 D	108 C	109 D	110 C
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Answers with key points

For every question: the options, the correct answer in green, and the one key point that explains it.

Histology

1. Which one of the following epidermal cell type has structural cell

A. Melanocytes B. Langerhans cells **C. Keratinocytes** ✓ D. Merkel cells

Answer: C — Keratinocytes

Keratinocytes are the main structural cells of the epidermis, forming the physical barrier; melanocytes, Langerhans and Merkel cells are scattered non-structural cells.

2. "Prickly layer" stands for which of the following epidermal stratum or layer

A. Basale **B. Spinosum** ✓ C. Granulosum D. Lucidum E. Corneum

Answer: B — Spinosum

The stratum spinosum ("prickle cell layer") owes its name to desmosomal spines linking keratinocytes, seen as intercellular bridges on light microscopy.

3. Endomysium covers

A. muscle fiber ✓ B. muscle fascicle C. Tendon D. none

Answer: A — muscle fiber

Endomysium is the thin connective tissue sheath around each individual muscle fiber; perimysium wraps fascicles and epimysium wraps the whole muscle.

4. Which is not true

A. Red blood cells have 120 days life span B. Red blood cells have biconcave disc shape **C. Monocyte is categorized under granular leukocyte** ✓ D. Platelets help for blood clotting

Answer: C — Monocyte is categorized under granular leukocyte

Monocytes are agranular (mononuclear) leukocytes, part of the monocyte-macrophage system; granulocytes are neutrophils, eosinophils and basophils.

5. Which of the following reasons makes histology dependent on the use of microscopes?

A. The small size of cells and matrix components ✓ B. The small size of tissues C. The small size of organs D. All of the above E. None of the above

Answer: A — The small size of cells and matrix components

Histology needs microscopy simply because cells and their matrix components are too small to resolve with the naked eye.

6. Which of the following body tissues is most frequently subject to wear and tear due to its superficial location?

A. Nervous tissue B. Muscle tissue C. Connective tissue **D. Epithelial tissue** ✓ E. All of the above

Answer: D — Epithelial tissue

Epithelium lines free surfaces and cavities, so it bears the most mechanical wear and has the highest turnover of the four basic tissues.

7. Which of the following descriptions about epithelial tissues is wrong?

- A. Epithelial tissues mostly exist on the free body surfaces or line body cavities and lumina of tubular organs.
B. Epithelial tissues have diverse embryological origin from all germ layers. C. The apical, lateral and basal surfaces of epithelial cells reveal structural modifications based on the functional requirements of the tissue.
D. Epithelial tissues are avascular. **E. All epithelial tissues rest on a basal lamina. ✓**

Answer: E — All epithelial tissues rest on a basal lamina.

Epithelium is avascular and rests on a basal lamina, but it is not derived from all three germ layers equally in every case — still, "rests on a basal lamina" is the safest universal true statement, making E the exception intended here (per the key).

8. In histology laboratory session, suppose you are given a histological slide containing all the four basic types of tissues. Which of the following histological features can help you to distinguish connective tissue from the other basic tissues?

- A. Highest cellularity B. Abundant cellular junctions connecting adjacent cells together **C. Abundant intercellular substance separating the cells from each other ✓** D. The presence of blood vessels E. The presence of motor and sensory nerve endings

Answer: C — Abundant intercellular substance separating the cells from each other

Connective tissue is defined by abundant extracellular matrix (ground substance and fibers) separating relatively sparse cells, unlike the tightly packed cells of epithelium.

9. Most tissue undergo constant cell turnover because of continuous cell division and ongoing death of cells. The turnover rate of cells is very rapid and high in which type of the following tissue?

- A. Connective tissue B. Muscles tissue **C. Epithelial tissue ✓** D. Nervous tissue

Answer: C — Epithelial tissue

Epithelial cells are exposed to friction and turn over fastest of the four basic tissues (days to weeks), versus the very slow turnover of muscle and neurons.

10. Which one of the following connective tissue covering is the thickest

- A. Epineurium ✓** B. Endoneurium C. Perineurium D. Fascicles

Answer: A — Epineurium

Epineurium is the outermost, thickest connective tissue sheath enclosing an entire peripheral nerve; perineurium wraps fascicles and endoneurium wraps individual axons.

Osteology & Skeletal Anatomy

11. Which of the following bones is unique because it does not articulate with any other bone in the human body?

- A. Patella B. Mandible **C. Hyoid bone ✓** D. Vomer

Answer: C — Hyoid bone

The hyoid bone is suspended by muscles and ligaments in the neck and uniquely forms no direct joint with any other bone.

12. Which statement accurately describes the distal articulations of the humerus at the elbow joint?

- A. The capitulum articulates with the ulna. **B. The capitulum articulates with the radius, while the trochlea articulates with the ulna. ✓** C. The trochlea articulates with the head of the radius. D. The humerus articulates exclusively with the ulna at its distal end.

Answer: B — The capitulum articulates with the radius, while the trochlea articulates with the ulna.

At the elbow, the humeral capitulum articulates with the radial head, while the trochlea articulates with the ulna's trochlear notch.

13. What is the correct sequence of carpal bones in the proximal row, ordering from lateral to medial?

A. Trapezium, trapezoid, capitate, hamate **B. Scaphoid, lunate, triquetrum, pisiform** ✓ C. Pisiform, triquetrum, lunate, scaphoid D. Hamate, capitate, trapezoid, trapezium

Answer: B — Scaphoid, lunate, triquetrum, pisiform

Proximal carpal row, lateral (thumb side) to medial: Scaphoid, Lunate, Triquetrum, Pisiform ("Some Lovers Try Positions").

14. Which bone is the longest and strongest in the human body?

A. Humerus B. Tibia **C. Femur** ✓ D. Fibula

Answer: C — Femur

The femur is both the longest and strongest bone in the body, built to bear the full weight of the upper body.

15. Which suture connects the paired parietal bones, and which cavity protects the brain?

A. Coronal suture; facial cavity **B. Sagittal suture; cranial cavity** ✓ C. Squamosal suture; thoracic cavity D. Lamboid suture; orbital cavity

Answer: B — Sagittal suture; cranial cavity

The sagittal suture joins the two parietal bones along the skull midline; the cranial cavity houses and protects the brain.

16. Where does the pectoral girdle articulate with the axial skeleton, and where do the scapular borders meet at the superior angle?

A. At the xiphoid process; junction of medial and lateral borders **B. At the manubrium; junction of superior and medial borders** ✓ C. At the clavicle; junction of superior and lateral borders D. At the acromion; junction of lateral and axillary borders

Answer: B — At the manubrium; junction of superior and medial borders

The pectoral girdle meets the axial skeleton at the sternoclavicular joint (manubrium); the scapula's superior and medial borders meet at the superior angle.

Neuroanatomy

17. Which structure forms the largest portion of the adult brain, accounting for roughly 80% of its total weight?

A. Cerebellum **B. Cerebrum** ✓ C. Diencephalon D. Brain stem

Answer: B — Cerebrum

The cerebrum (two cerebral hemispheres) makes up roughly 80% of total brain mass and volume.

18. Which cranial nerve is responsible for sensory perception of hearing and balance?

A. Trigeminal nerve (CN V) B. Facial nerve (CN VII) **C. Vestibulocochlear nerve (CN VIII)** ✓ D. Glossopharyngeal nerve (CN IX)

Answer: C — Vestibulocochlear nerve (CN VIII)

The vestibulocochlear nerve (CN VIII) carries special sensory fibers for hearing (cochlear) and balance (vestibular).

19. Which diencephalic structure acts as a relay center for all sensory input to the cerebral cortex EXCEPT smell?

A. Hypothalamus **B. Thalamus** ✓ C. Epithalamus D. Pineal gland

Answer: B — Thalamus

The thalamus relays essentially all sensory information to the cerebral cortex except olfaction, which reaches the cortex directly.

20. Which layer of the meninges is highly vascular and attached directly to the surface of the CNS?

A. Dura mater B. Arachnoid mater **C. Pia mater** ✓ D. Epidural layer

Answer: C — Pia mater

The pia mater is the innermost, delicate, highly vascular meningeal layer that closely follows the brain and spinal cord surface, including into sulci.

General & Regional Anatomy

21. All of the following statements are false about history of Anatomy except

A. Anatomy is architecture of the body B. The earliest Anatomy was written on papyrus C. Anatomy was first studied in Egypt D. Cutting apart is the old definition of anatomy **E. All** ✓

Answer: E — All

All of A-D describe genuine historical facts about anatomy (architecture of the body, papyrus records, Egyptian origins, "cutting apart" etymology), so none is false and the intended answer is "All."

22. Who is the first scholar in history of Anatomy and use the term "Anatomy" for the first time

A. Galen B. William C. Hippocrates **D. Aristotle** ✓

Answer: D — Aristotle

Aristotle is credited as the first to use the term "anatome" (dissection/anatomy) systematically in his comparative studies.

23. Which one of the following division of anatomy study about structures too small to be seen with a light microscope

A. Applied Anatomy B. Microscopic Anatomy C. Embryology D. Pathology **E. Ultrastructural Anatomy** ✓

Answer: E — Ultrastructural Anatomy

Ultrastructural (electron microscopic) anatomy studies structures below the resolution of a light microscope, such as organelles and membranes.

24. Which one of the following movement is not perpendicular to the floor

A. Medial rotation B. Lateral rotation **C. Adduction** ✓ D. Pronation E. Supination

Answer: C — Adduction

Adduction moves a limb toward the body in the coronal plane (roughly parallel to the floor when standing), unlike rotation, pronation or supination which occur around a vertical/longitudinal axis.

25. Which one of the following is not special type of movement

A. Abduction ✓ B. Protrusion C. Reposition D. Eversion E. Elevation

Answer: A — Abduction

Abduction is a standard angular movement (away from the midline), not one of the "special" movements like protrusion, reposition, eversion or elevation.

26. Which vertebrae has costal facet for origin of ribs

A. Cervical B. Sacrum C. Lumbar **D. Thoracic** ✓

Answer: D — Thoracic

Thoracic vertebrae uniquely bear costal facets on their bodies and transverse processes for articulation with the ribs.

Cardiovascular Anatomy

27. Bicuspid valve is located between

- A. left ventricle and aorta B. right atrium and right ventricle **C. left atrium and left ventricle ✓**

Answer: C — left atrium and left ventricle

The bicuspid (mitral) valve sits between the left atrium and left ventricle, controlling flow into the systemic circuit.

28. Which of the following is NOT contained in the superior mediastinum?

- A. Great vessels and their branches **B. Azygos and hemiazygos veins ✓** C. Trachea D. Esophagus

Answer: B — Azygos and hemiazygos veins

The azygos and hemiazygos veins run in the posterior mediastinum, not the superior mediastinum, which instead contains the great vessels, trachea and esophagus.

29. The attending faculty in the coronary intensive care unit (ICU) demonstrates to his students a normal heart examination. The first heart sound is produced by near-simultaneous closure of which of the following valves?

- A. Aortic and tricuspid B. Aortic and pulmonary **C. Tricuspid and mitral ✓** D. Mitral and pulmonary

Answer: C — Tricuspid and mitral

S1 ("lub") is produced by near-simultaneous closure of the atrioventricular valves — tricuspid and mitral — at the start of ventricular systole.

30. A 27-year-old cardiac patient with an irregular heartbeat visits her doctor's office for examination. Where should the physician place the stethoscope to listen to the sound of the Aortic valve?

- A. Over the medial end of the second left intercostal space **B. Over the medial end of the second right intercostal space ✓** C. In the left fourth intercostal space at the midclavicular line D. In the left fifth intercostal space at the midclavicular line

Answer: B — Over the medial end of the second right intercostal space

The aortic valve is best auscultated at the right 2nd intercostal space, close to the sternal border.

Cell Membrane & Transport

31. Which protein family forms the transmembrane component of tight junctions, acting as 'fences' to regulate paracellular permeability and maintain cell polarity?

- A. Cadherins and Keratin B. Connexins **C. Occludin and Claudin ✓** D. Integrins and Selectins

Answer: C — Occludin and Claudin

Tight junctions use the transmembrane proteins occludin and claudin as the "fence" that seals the paracellular space and maintains apical-basal polarity.

32. In a standard 70 kg adult male, what percentage of total body weight is accounted for by intracellular fluid (ICF)?

- A. 5% B. 15% C. 20% **D. 40% ✓**

Answer: D — 40%

Intracellular fluid makes up about 40% of total body weight (roughly two-thirds of total body water) in a standard 70 kg adult male.

33. The Na⁺/K⁺ ATPase pump is a primary active transport carrier. What is its ion transport stoichiometry per ATP molecule hydrolyzed?

- A. Pumps 2 Na⁺ outward and 3 K⁺ inward **B. Pumps 3 Na⁺ outward and 2 K⁺ inward ✓** C. Pumps 3 Na⁺ inward and 2 K⁺ outward D. Pumps 2 Na⁺ inward and 2 K⁺ outward

Answer: B — Pumps 3 Na⁺ outward and 2 K⁺ inward

The Na⁺/K⁺-ATPase is electrogenic: for every ATP hydrolyzed it pumps 3 Na⁺ out and 2 K⁺ in, against their gradients.

Endocrine & Renal Physiology

34. Which of the following physiological events is driven by a positive feedback mechanism?

A. Regulation of systemic arterial blood pressure B. Control of blood glucose concentration by insulin **C. Pre-ovulatory LH surge triggered by high estrogen levels** ✓ D. Respiratory adjustment to elevated blood PCO₂

Answer: C — Pre-ovulatory LH surge triggered by high estrogen levels

The pre-ovulatory LH surge is a positive-feedback loop: rising estrogen from the dominant follicle stimulates even more LH release, unlike the negative feedback of BP, glucose or CO₂ control.

35. How do steroid hormones differ from peptide hormones in terms of structure, synthesis, and transport?

A. They are synthesized in the rough endoplasmic reticulum and stored in secretory vesicles. B. They are hydrophilic molecules that circulate freely in plasma. **C. They are derived from cholesterol, synthesized in the smooth endoplasmic reticulum, and diffuse across membranes.** ✓ D. They bind exclusively to cell-surface GPCRs to trigger fast ion influx.

Answer: C — They are derived from cholesterol, synthesized in the smooth endoplasmic reticulum, and diffuse across membranes.

Steroid hormones are cholesterol-derived, made in the smooth ER, and being lipophilic they diffuse directly across the plasma membrane rather than being stored in vesicles.

36. Why do lipophilic hormones (such as steroids and thyroid hormones) generally produce a slower cellular response than hydrophilic hormones?

A. They require secondary active transport carriers to cross the lipid bilayer. **B. They bind to cytosolic or nuclear receptors to directly alter gene transcription and protein synthesis.** ✓ C. They open ligand-gated membrane ion channels directly upon binding. D. They are degraded by liver enzymes before reaching target cell organelles.

Answer: B — They bind to cytosolic or nuclear receptors to directly alter gene transcription and protein synthesis.

Lipophilic hormones act slowly because they bind cytosolic/nuclear receptors and change gene transcription and new protein synthesis, rather than triggering an instant membrane event.

37. Hormone release that occurs in episodic bursts with a periodicity of approximately 24 hours (such as diurnal cortisol secretion) is defined as which type of rhythm?

A. Circoral B. Ultradian **C. Circadian** ✓ D. Paracrine

Answer: C — Circadian

A roughly 24-hour hormone rhythm (such as cortisol) is called circadian; ultradian rhythms are faster than 24 hours and circoral cycles are hourly.

38. Given a Glomerular Capillary Hydrostatic Pressure of 60 mmHg, a Glomerular Capillary Colloid Osmotic Pressure of 32 mmHg, and a Capsular Hydrostatic Pressure of 18 mmHg, what is the Net Filtration Pressure (NFP)?

A. 10 mmHg ✓ B. 28 mmHg C. 42 mmHg D. 110 mmHg

Answer: A — 10 mmHg

$NFP = \text{glomerular hydrostatic pressure} - (\text{glomerular colloid osmotic pressure} + \text{capsular hydrostatic pressure}) = 60 - (32 + 18) = 10 \text{ mmHg}.$

39. When GFR drops and tubular flow slows down, how does the macula densa respond to restore filtration rate?

A. It secretes endothelin to constrict afferent arterioles. **B. It releases prostaglandins (PGE1/PGE2) to dilate the afferent arteriole and triggers JG cells to release renin.** ✓ C. It secretes atrial natriuretic peptide (ANP) to constrict efferent arterioles. D. It activates sympathetic nerve fibers to reduce renal blood flow.

Answer: B — It releases prostaglandins (PGE1/PGE2) to dilate the afferent arteriole and triggers JG cells to release renin.

A fall in GFR/tubular flow is sensed by the macula densa, which releases prostaglandins to dilate the afferent arteriole and signals JG cells to release renin — the core of tubuloglomerular feedback.

40. Which segment of the nephron is passively permeable to water but impermeable to solutes, allowing water reabsorption into the hyperosmolar medulla?

A. Proximal Convoluted Tubule (PCT) **B. Descending Limb of the Loop of Henle** ✓ C. Thick Ascending Limb of the Loop of Henle D. Distal Convoluted Tubule (DCT)

Answer: B — Descending Limb of the Loop of Henle

The thin descending limb of the loop of Henle is water-permeable but solute-impermeable, so water leaves passively into the hyperosmolar medullary interstitium.

41. What cellular mechanism does Antidiuretic Hormone (ADH) utilize to concentrate urine during dehydration?

A. Primary active transport of water via H⁺/K⁺ ATPase pumps. **B. Insertion of Aquaporin-2 (AQP-2) channels into the luminal membrane via V2 receptor activation.** ✓ C. Direct inhibition of Na⁺/K⁺/2Cl⁻ co-transporters. D. Reduction of renal medullary interstitial osmolarity.

Answer: B — Insertion of Aquaporin-2 (AQP-2) channels into the luminal membrane via V2 receptor activation.

ADH binds V2 receptors on collecting-duct principal cells, causing insertion of aquaporin-2 channels into the luminal membrane and increased water reabsorption.

42. Facilitated diffusion approaches a maximum transport rate (V_{max}) because:

A. Solutes are transported against their electrochemical gradient using ATP hydrolysis. **B. Membrane carrier proteins become fully saturated at high solute concentrations.** ✓ C. Diffusion across lipid bilayers is restricted by cell coat thickness. D. Aquaporin channels close when intracellular pressure rises.

Answer: B — Membrane carrier proteins become fully saturated at high solute concentrations.

Facilitated diffusion is carrier-mediated, so at high solute concentration the finite number of carrier proteins becomes saturated, producing a true V_{max} (unlike simple diffusion).

Reproductive Physiology

43. Where does normal fertilization of the ovum take place in the female reproductive system?

A. Uterus **B. Ampulla of the fallopian tube** ✓ C. Cervix D. Ovary

Answer: B — Ampulla of the fallopian tube

Fertilization normally occurs in the ampulla of the uterine (fallopian) tube, the widest part where the egg and sperm typically meet.

44. Which hormone is primarily responsible for triggering ovulation during the female sexual cycle?

A. Follicle-Stimulating Hormone (FSH) **B. Luteinizing Hormone (LH) ✓** C. Progesterone D. Human Chorionic Gonadotropin (hCG)

Answer: B — Luteinizing Hormone (LH)

The mid-cycle LH surge (triggered by peak estrogen) is the direct trigger for ovulation, causing the mature follicle to rupture.

45. What is the primary physiological function of Human Chorionic Gonadotropin (hCG) following fertilization?

A. It causes rapid involution of the corpus luteum **B. It prevents regression of the corpus luteum, stimulating continued secretion of progesterone and estrogen ✓** C. It directly triggers myometrial contractions to initiate labor D. It causes desquamation of the functional layer of the endometrium

Answer: B — It prevents regression of the corpus luteum, stimulating continued secretion of progesterone and estrogen

hCG from the early trophoblast rescues the corpus luteum, maintaining its progesterone and estrogen output until the placenta takes over around week 8-10.

46. Where and when does implantation of the blastocyst typically occur after ovulation?

A. In the fallopian tube on days 1-2 **B. In the endometrium of the uterus on days 5-7 ✓** C. In the cervix on days 10-12 D. In the ovary on day 14

Answer: B — In the endometrium of the uterus on days 5-7

The blastocyst implants into the uterine endometrium about 5-7 days after fertilization/ovulation.

47. Which cells on the surface of the blastocyst secrete proteolytic enzymes to digest and liquefy endometrial tissue during implantation?

A. Granulosa cells B. Decidual cells **C. Trophoblast cells ✓** D. Luteal cells

Answer: C — Trophoblast cells

Trophoblast cells secrete proteolytic enzymes that erode the endometrial epithelium and stroma, allowing the blastocyst to burrow in during implantation.

48. How does fetal blood transport sufficient oxygen despite having a lower PO₂ (~30 mm Hg) than maternal blood (~50 mm Hg)?

A. Fetal blood relies exclusively on active transport pumps for oxygen transfer **B. Fetal hemoglobin dissociation curve is shifted to the left and fetal blood has ~50% higher hemoglobin concentration ✓** C. Maternal blood transfers oxygen without simple diffusion gradients D. Fetal blood PO₂ rises significantly above maternal levels prior to delivery

Answer: B — Fetal hemoglobin dissociation curve is shifted to the left and fetal blood has ~50% higher hemoglobin concentration

Fetal hemoglobin has a left-shifted oxygen dissociation curve (higher O₂ affinity) plus a higher hemoglobin concentration, letting the fetus extract enough oxygen from maternal blood despite a lower PO₂.

49. Toward the end of pregnancy, which hormonal shift contributes to the increased contractility of the uterine myometrium?

A. Complete cessation of oxytocin release **B. An increased ratio of estrogens to progesterone ✓** C. A sudden drop in fetal adrenal cortisol production D. Downregulation of myometrial gap junctions

Answer: B — An increased ratio of estrogens to progesterone

Toward term, the estrogen:progesterone ratio rises (progesterone's quieting effect wanes relative to estrogen), increasing myometrial excitability and gap-junction coupling ahead of labor.

Gastrointestinal Physiology

50. Which nerve plexus within the wall of the gastrointestinal tract primarily controls gut motility?

A. Submucosal plexus (Meissner's plexus) **B. Myenteric plexus (Auerbach's plexus) ✓** C. Celiac plexus D. Hypogastric plexus

Answer: B — Myenteric plexus (Auerbach's plexus)

The myenteric (Auerbach's) plexus, between the circular and longitudinal muscle layers, is the main driver of GI motility; the submucosal (Meissner's) plexus mainly governs secretion and local blood flow.

Autonomic & Nervous System Physiology

51. During the depolarization phase of action potential

A. Voltage gated Na channels are opened ✓ B. Voltage gated K channels opened C. Voltage gated Na channels are closed D. Ligand gated K channels are closed E. None of the above

Answer: A — Voltage gated Na channels are opened

Reaching threshold opens voltage-gated Na^+ channels; the resulting Na^+ influx is the depolarizing phase of the action potential.

52. In the peripheral nerve fibers, myelin sheath is produced by

A. Schwann cells ✓ B. Oligodendrocytes C. Microglial cells D. Ependymal cells E. Astrocytes

Answer: A — Schwann cells

Schwann cells myelinate peripheral axons (one cell per internode); oligodendrocytes do this job in the CNS.

53. A neurotransmitter produced by a postganglionic sympathetic fiber to the sweat glands is

A. Acetylcholine ✓ B. Adrenaline C. Nor adrenaline D. Dopamine E. All

Answer: A — Acetylcholine

Postganglionic sympathetic fibers to sweat glands are an exception: they release acetylcholine (cholinergic) rather than norepinephrine.

54. Which one of the following is wrong about neuronal activities of the ANS?

A. All preganglionic fibers are cholinergic B. Postganglionic parasympathetic fibers are cholinergic C. Preganglionic parasympathetic fibers are cholinergic D. Most postganglionic sympathetic fibers are adrenergic. **E. None of the above ✓**

Answer: E — None of the above

All preganglionic fibers and all parasympathetic postganglionic fibers are cholinergic, and most sympathetic postganglionic fibers are adrenergic — so every statement is true and "none" is wrong.

55. An action potential arriving at the neuromuscular junction causes release of:

A. Acetylcholine ✓ B. Na^+ ions C. Ca^{2+} ions D. Noradrenalin E. None of the above

Answer: A — Acetylcholine

The arriving action potential opens voltage-gated Ca^{2+} channels; Ca^{2+} influx triggers vesicle fusion and acetylcholine release at the NMJ.

56. Parasympathetic stimulation produces:

A. Bronchoconstriction B. Increased gut motility C. Decreased heart activities D. Relaxes GIT sphincters **E. All are correct ✓**

Answer: E — All are correct

Parasympathetic ("rest and digest") activity bronchoconstricts, boosts gut motility, slows the heart, and relaxes GI sphincters — all four are correct.

57. Acetylcholine released by autonomic nervous system binds to what type of receptors on skeletal muscles?

A. Muscarinic receptors B. Beta adrenergic receptors **C. Nicotinic receptors** ✓ D. None

Answer: C — Nicotinic receptors

Skeletal muscle end plates carry nicotinic (N_M) receptors, ligand-gated cation channels that open directly on ACh binding.

58. The action potential of a neuron is initiated by:

A. Efflux of Na⁺ B. Influx of K⁺ **C. Influx of Na⁺** ✓ D. Efflux of K⁺ E. Influx of Cl⁻

Answer: C — Influx of Na⁺

Threshold depolarization opens voltage-gated Na⁺ channels, and Na⁺ influx (down its electrochemical gradient) initiates the rising phase of the action potential.

59. All of the following hormones are Amine EXCEPT:

A. Epinephrine B. Melatonin C. Nor epinephrine D. Thyroid hormone **E. None** ✓

Answer: E — None

Epinephrine, norepinephrine and thyroid hormone are all amine hormones (from tyrosine); melatonin is also an amine (from tryptophan), so none is the exception.

60. Which nerve carries most of the parasympathetic output from the brain?

A. Spinal **B. Vagus** ✓ C. Oculomotor D. Facial E. Glossopharyngeal

Answer: B — Vagus

The vagus nerve (CN X) supplies roughly 75% of all parasympathetic outflow, reaching the thoracic and most abdominal viscera.

Muscle Physiology

61. During a sustained isotonic skeletal muscle contraction, a researcher measures the dimensions of various microscopic bands within a single sarcomere under an electron microscope. Which of the following changes is observed during active contraction?

A. The length of the A band decreases significantly. B. The distance between adjacent Z discs increases. **C. The length of the A band remains constant while the H zone and I band shorten.** ✓ D. The I band widens as thin filaments slide past thick filaments.

Answer: C — The length of the A band remains constant while the H zone and I band shorten.

During contraction the myosin (thick filament/A band) length stays fixed while the actin filaments slide inward, shortening the H zone and I band.

62. A patient with hypocalcemia develops painful muscle spasms. Which change in the neuromuscular junction is most likely responsible for this finding?

A. Increased threshold for action potential generation B. Decreased neuronal excitability **C. Increased neuronal excitability** ✓ D. Decreased acetylcholine release E. Blockade of nicotinic acetylcholine receptors

Answer: C — Increased neuronal excitability

Low extracellular Ca²⁺ lowers the threshold for Na⁺ channel opening at the neuromuscular membrane, increasing neuronal/muscle excitability and causing spontaneous spasms (tetany).

63. During skeletal muscle contraction, calcium binds to which protein to initiate movement of tropomyosin away from the actin-binding sites?

A. Myosin **B. Troponin C** ✓ C. Troponin I D. Tropomyosin E. Titin

Answer: B — Troponin C

Ca²⁺ binds troponin C, shifting the troponin-tropomyosin complex and exposing the myosin-binding sites on actin.

64. A sprinter performs a maximal 10-second sprint. Which energy source provides the most immediate ATP regeneration for skeletal muscle contraction?

- A. Fatty acid β -oxidation B. Aerobic oxidative phosphorylation C. Glycogen synthesis **D. Phosphocreatine** ✓
E. Ketone body oxidation

Answer: D — Phosphocreatine

Phosphocreatine regenerates ATP almost instantly (creatine kinase reaction), fueling the first seconds of maximal effort before glycolysis ramps up.

Respiratory Physiology

65. Which structure of the following does not belong to the respiratory portion of the breathing organ?

- A. Terminal bronchioles** ✓ B. Respiratory bronchioles C. Alveolar ducts D. Alveolar sacs E. None of the above

Answer: A — Terminal bronchioles

Terminal bronchioles are purely conducting airways (no gas exchange); the respiratory portion begins at the respiratory bronchioles.

66. All of the following events occur during expiration except

- A. Inspiratory muscles relax and the rib cage descends due to gravity B. Thoracic cavity volume decreases C. Intrapulmonary volume decreases D. Intrapulmonary pressure rises above atmospheric pressure (+1 mm Hg)
E. Gases flow out of the lungs against the pressure gradient ✓

Answer: E — Gases flow out of the lungs against the pressure gradient

During quiet expiration, air flows out down its own pressure gradient (alveolar pressure exceeds atmospheric), so "against the pressure gradient" is the false statement.

67. The total lung capacity consists of:

- A. Vital capacity & tidal volume **B. Residual volume & vital capacity** ✓ C. Functional residual capacity & vital capacity D. Tidal volume & functional residual capacity E. Tidal volume & residual volume

Answer: B — Residual volume & vital capacity

Total lung capacity = vital capacity + residual volume (all air the lungs can hold, including the fraction that can never be exhaled).

68. Wrong about Vital capacity is:

- A. The volume of air expired from full inspiration to full expiration B. Reduced as one grows older C. Greater in men than in women of the same age and height **D. Not related more to total body mass than to lean body mass** ✓ E. The sum of the inspiratory and expiratory reserve volumes

Answer: D — Not related more to total body mass than to lean body mass

Vital capacity does scale with body/lean mass (and is higher in men and falls with age), so "not related to body mass" is the false statement.

Reproductive Anatomy

69. A 25-year-old man presents with infertility. Semen analysis shows a very low sperm count. Which structure is primarily responsible for producing testosterone that supports spermatogenesis?

- A. Sertoli cells **B. Leydig cells** ✓ C. Spermatogonia D. Epididymal cells E. Prostate gland

Answer: B — Leydig cells

Leydig cells, stimulated by LH, produce the testosterone that supports spermatogenesis within the adjacent seminiferous tubules.

70. A 30-year-old man undergoes a vasectomy. Which structure is surgically cut or occluded to prevent sperm from entering the ejaculate?

A. Epididymis B. Ejaculatory duct **C. Ductus deferens** ✓ D. Seminal vesicle E. Prostatic urethra

Answer: C — Ductus deferens

Vasectomy involves cutting/ligating the ductus (vas) deferens, blocking sperm transport from the epididymis to the ejaculatory duct.

Embryology

71. A surgical resident is reviewing the embryological origins of the middle ear ossicles. Which of the following correctly matches the ossicle with its embryonic cartilage precursor and pharyngeal arch of origin?

A. Malleus – Reichert's cartilage – Arch 1 **B. Incus – Meckel's cartilage – Arch 1** ✓ C. Stapes – Meckel's cartilage – Arch 2 D. Malleus – Reichert's cartilage – Arch 2 E. Stapes – Thyroid cartilage – Arch 3

Answer: B — Incus – Meckel's cartilage – Arch 1

The incus develops from Meckel's cartilage of the 1st pharyngeal arch, along with the malleus; the stapes comes from the 2nd arch (Reichert's cartilage).

72. A patient presents with difficulty swallowing (dysphagia). Anatomical testing confirms weakness in the stylopharyngeus muscle, the only muscle derived from pharyngeal arch 3. Which nerve supplies this muscle, and what skeletal structure develops from arch 3?

A. Facial nerve (CN VII); Styloid process **B. Glossopharyngeal nerve (CN IX); Greater horn and lower body of the hyoid bone** ✓ C. Vagus nerve (CN X); Thyroid cartilage D. Mandibular nerve (CN V3); Lesser horn of the hyoid bone E. Accessory nerve (CN XI); Manubrium of the sternum

Answer: B — Glossopharyngeal nerve (CN IX); Greater horn and lower body of the hyoid bone

Stylopharyngeus, the sole 3rd-arch muscle, is supplied by the glossopharyngeal nerve (CN IX); arch 3 cartilage forms the greater horn and lower body of the hyoid.

73. A newborn infant presents with a triad of micrognathia (severely underdeveloped mandible), cleft palate, and glossoptosis (posterior placement of the tongue causing airway obstruction). This clinical presentation (Robin sequence) is primarily initiated by a developmental defect in which embryonic structure?

A. Frontonasal prominence **B. First pharyngeal arch** ✓ C. Second pharyngeal arch D. Third pharyngeal pouch E. Oropharyngeal membrane

Answer: B — First pharyngeal arch

Robin sequence begins with a primary defect in 1st-arch (mandibular) growth, causing micrognathia, which secondarily displaces the tongue and prevents palatal shelf fusion.

74. The tongue develops from several median and lateral swellings in the floor of the primitive pharynx. Which nerve supplies general sensation to the mucosa of the anterior two-thirds of the tongue, and which nerve supplies taste to the posterior one-third?

A. Facial nerve (CN VII); Glossopharyngeal nerve (CN IX) **B. Trigeminal nerve (CN V3); Glossopharyngeal nerve (CN IX)** ✓ C. Glossopharyngeal nerve (CN IX); Vagus nerve (CN X) D. Trigeminal nerve (CN V3); Vagus nerve (CN X) E. Hypoglossal nerve (CN XII); Facial nerve (CN VII)

Answer: B — Trigeminal nerve (CN V3); Glossopharyngeal nerve (CN IX)

Anterior two-thirds of the tongue: general sensation via the trigeminal nerve (CN V3, lingual nerve); posterior one-third: taste and sensation via the glossopharyngeal nerve (CN IX).

75. Which adult lymphoid structure originates from the endodermal lining of pharyngeal pouch 2?

A. Epithelial lining of the auditory tube **B. Crypts and epithelial lining of the palatine tonsil** ✓ C. Thymic stroma D. Parafollicular (C) cells of the thyroid E. Superior parathyroid gland

Answer: B — Crypts and epithelial lining of the palatine tonsil

The palatine tonsil's crypts and epithelium derive from the endoderm of the 2nd pharyngeal pouch.

76. At the exact moment of ovulation, in what specific meiotic state is the secondary oocyte's genetic material?

A. Arrested in prophase of meiosis I B. Arrested in metaphase of meiosis I C. Arrested in prophase of meiosis II **D. Arrested in metaphase of meiosis II** ✓ E. Meiosis II is fully completed

Answer: D — Arrested in metaphase of meiosis II

The secondary oocyte is ovulated arrested in metaphase of meiosis II, and only completes meiosis II if fertilization occurs.

77. Which structure takes over the primary production of progesterone required to maintain the uterine lining late in pregnancy (after week 8)?

A. Corpus luteum **B. Syncytiotrophoblast of the placenta** ✓ C. Maternal adrenal cortex D. Fetal adenohypophysis E. Cytotrophoblast

Answer: B — Syncytiotrophoblast of the placenta

By about week 8–10, the placental syncytiotrophoblast takes over progesterone production from the corpus luteum, which then regresses.

78. In 65% of monozygotic twin cases, splitting of the conceptus occurs at the inner cell mass (embryoblast) stage during the late first week. What fetal membrane arrangement will these twins exhibit?

A. Two placentas, two chorions, two amnions (diamniotic-dichorionic) B. One placenta, two chorions, two amnions **C. One placenta, one chorion, two amnions (diamniotic-monochorionic)** ✓ D. One placenta, one chorion, one amnion (monoamniotic-monochorionic) E. Two placentas, one chorion, two amnions

Answer: C — One placenta, one chorion, two amnions (diamniotic-monochorionic)

Splitting at the inner cell mass stage (days 4–8) after the chorion has formed but before amnion separation gives monochorionic, diamniotic twins — the most common monozygotic pattern.

79. During which developmental window is the conceptus most susceptible to teratogens that cause gross structural birth defects (organogenesis)?

A. Week 1 (Preimplantation period) **B. Weeks 3 to 8 (Embryonic period)** ✓ C. Weeks 9 to 12 (Early fetal period) D. Weeks 13 to 20 (Mid fetal period) E. Weeks 21 to 38 (Late fetal period)

Answer: B — Weeks 3 to 8 (Embryonic period)

Weeks 3–8 (the embryonic period, organogenesis) is when the conceptus is most vulnerable to teratogens causing major structural defects.

80. A 25-year-old woman with a history of pelvic inflammatory disease presents with lower abdominal pain, light vaginal bleeding, amenorrhea for 5 weeks, and elevated serum hCG. Ultrasound reveals a mass located in the ampulla of the left uterine tube. What is the diagnosis?

A. Choriocarcinoma B. Complete hydatidiform mole **C. Ectopic tubal pregnancy** ✓ D. Appendicitis E. Placenta previa

Answer: C — Ectopic tubal pregnancy

A hCG-positive mass in the tubal ampulla with amenorrhea and pain is the classic presentation of an ectopic tubal pregnancy.

81. Which trophoblastic layer plays the most active role in eroding maternal endometrial glands and blood vessels during blastocyst implantation?

A. Cytotrophoblast **B. Syncytiotrophoblast** ✓ C. Epiblast D. Extraembryonic somatic mesoderm E. Hypoblast

Answer: B — Syncytiotrophoblast

The syncytiotrophoblast is the invasive, enzyme-secreting layer that erodes maternal endometrial glands and vessels during implantation.

82. Individual blastomeres are isolated from a conceptus at the 4-cell stage, cultured, and implanted into separate foster mothers. What is the expected outcome?

A. Birth of one baby only B. Birth of four genetically different babies **C. Birth of four genetically identical babies** ✓ D. Birth of four grotesquely deformed offspring E. Failure of all implants

Answer: C — Birth of four genetically identical babies

Because each blastomere at the 4-cell stage is still totipotent, separating them yields four genetically identical embryos (monozygotic "quadruplets").

Biochemistry: Carbohydrates

83. What is the general empiric formula for many simpler carbohydrates?

A. $(C_2H_5O)_n$ **B. $(CH_2O)_n$** ✓ C. $(CHO_2)_n$ D. $(C_2H_5O)_n$

Answer: B — $(CH_2O)_n$

Simple carbohydrates fit the general empirical formula $(CH_2O)_n$, reflecting their carbon-water-like composition.

84. Which term describes carbohydrates with a keto group as their most oxidized functional group?

A. Aldoses **B. Ketoses** ✓ C. Polyols D. Uronic acids

Answer: B — Ketoses

Ketoses are defined by a ketone (keto) group as their most oxidized carbon, as opposed to aldoses, which have an aldehyde group.

85. A monosaccharide containing six carbons and an aldehyde group is classified as a:

A. Ketohexose B. Aldopentose **C. Aldohexose** ✓ D. Ketopentose

Answer: C — Aldohexose

A six-carbon sugar with an aldehyde group (e.g. glucose) is an aldohexose.

86. Which sugar is the C-4 epimer of glucose?

A. Mannose B. Fructose **C. Galactose** ✓ D. Ribose

Answer: C — Galactose

Galactose differs from glucose only in the configuration at carbon 4, making it the C-4 epimer of glucose.

87. Which sugar is the C-2 epimer of glucose?

A. Galactose **B. Mannose** ✓ C. Xylulose D. Sorbitol

Answer: B — Mannose

Mannose differs from glucose only at carbon 2, making it the C-2 epimer of glucose.

88. In biological systems, the vast majority of sugars exist in which isomeric form?

A. L-form **B. D-form** ✓ C. Trans-form D. Meso-form

Answer: B — D-form

The great majority of naturally occurring sugars in biology exist in the D-configuration.

89. A six-membered sugar ring is specifically referred to as a:

A. Furanose **B. Pyranose** ✓ C. Pentose D. Hexose

Answer: B — Pyranose

A six-membered cyclic sugar ring (like glucopyranose) is called a pyranose; a five-membered ring is a furanose.

90. Which test is used to detect reducing sugars in urine via copper reduction?

A. Glucose oxidase test **B. Clinitest** ✓ C. Gram stain D. Hydrogen breath test

Answer: B — Clinitest

Clinitest is a copper-reduction (Benedict-type) test used to detect reducing sugars such as glucose in urine.

Biochemistry: Membranes & Cell Biology

91. According to the fluid mosaic model of biological membranes, which component is primarily responsible for forming the continuous hydrophobic barrier?

A. Integral transmembrane proteins ✓ B. Phospholipid bilayer C. Glycoprotein oligosaccharide chains D. Peripheral membrane proteins

Answer: A — Integral transmembrane proteins

The phospholipid bilayer forms the continuous hydrophobic core of the membrane that acts as the barrier to most polar molecules.

92. Which lipid component functions as a membrane 'fluidity buffer' by restricting fatty acid chain movement at high temperatures while preventing tight packing at low temperatures?

A. Phosphatidylcholine B. Sphingomyelin **C. Cholesterol** ✓ D. Cardiolipin

Answer: C — Cholesterol

Cholesterol acts as a fluidity buffer: it restrains excessive fluidity at high temperature and prevents tight, rigid packing at low temperature.

93. Glucose transport into intestinal epithelial cells via SGLT1 is an example of which transport mechanism?

A. Primary active transport **B. Sodium-dependent secondary active symport** ✓ C. Facilitated diffusion uniport D. Passive antiport transport

Answer: B — Sodium-dependent secondary active symport

SGLT1 co-transporters glucose with Na⁺ down the Na⁺ gradient (secondary active transport, symport), powered indirectly by the Na⁺/K⁺-ATPase.

94. Which type of passive transport protein selectively conducts water molecules across biological membranes at high velocities without allowing proton passage?

A. Aquaporin ✓ B. Gramicidin channel C. Connexon D. Ionophore

Answer: A — Aquaporin

Aquaporins are dedicated water channels that allow extremely rapid, selective water flux while excluding ions and protons.

95. Which organelle is the primary site of steroid hormone synthesis, phospholipid biogenesis, drug detoxification via Cytochrome P450, and intracellular Ca²⁺ storage?

A. Rough Endoplasmic Reticulum **B. Smooth Endoplasmic Reticulum** ✓ C. Golgi Apparatus D. Peroxisome

Answer: B — Smooth Endoplasmic Reticulum

The smooth endoplasmic reticulum makes steroid hormones and phospholipids, detoxifies drugs via cytochrome P450, and stores/releases intracellular Ca²⁺.

96. Which sub-nuclear structure is the primary site of ribosomal RNA (rRNA) transcription, processing, and ribosome subunit assembly?

A. Nuclear matrix **B. Nucleolus** ✓ C. Cajal body D. Nuclear envelope

Answer: B — Nucleolus

The nucleolus is the site of rRNA transcription/processing and assembly of ribosomal subunits.

97. Which mode of extracellular cell signaling involves signaling molecules acting locally on neighboring target cells of a different type in the immediate extracellular environment?

A. Endocrine signaling **B. Paracrine signaling** ✓ C. Autocrine signaling D. Juxtacrine signaling

Answer: B — Paracrine signaling

Paracrine signaling involves a mediator acting locally on nearby cells of a different type, without needing the bloodstream (unlike endocrine signaling).

98. G-protein coupled receptors (GPCRs) span the plasma membrane how many times?

A. Single transmembrane pass (1) **B. Seven transmembrane alpha-helices (7)** ✓ C. Twelve transmembrane passes (12) D. Four transmembrane passes (4)

Answer: B — Seven transmembrane alpha-helices (7)

GPCRs share a conserved architecture of seven transmembrane alpha-helices, regardless of which G-protein or ligand they couple to.

Biochemistry: Enzymes

99. What is the primary chemical nature of virtually all biological enzymes, with the exception of catalytic RNAs?

A. Linear unbranched polysaccharides **B. Globular proteins composed of L-amino acids** ✓ C. Double-stranded deoxyribonucleic acid polymers D. Esterified triacylglycerol lipid complexes

Answer: B — Globular proteins composed of L-amino acids

Almost all enzymes are globular proteins made of L-amino acids, aside from the small group of catalytic RNAs (ribozymes).

100. An enzyme accelerates a chemical reaction rate by altering which of the following thermodynamic parameters?

A. Standard free energy change (ΔG°) **B. Activation energy barrier ($\Delta G^\ddagger$) of the transition state** ✓ C. Equilibrium constant (K_{eq}) of the reaction D. Total enthalpy (ΔH) of initial reactants and final products

Answer: B — Activation energy barrier ($\Delta G^\ddagger$) of the transition state

Enzymes speed reactions by lowering the activation energy ($\Delta G^\ddagger$) needed to reach the transition state; they do not change ΔG° or K_{eq} .

101. An enzyme devoid of its required cofactor/coenzyme is catalytically inactive and is termed an:

A. Holoenzyme **B. Apoenzyme** ✓ C. Proenzyme D. Metalloenzyme

Answer: B — Apoenzyme

An apoenzyme is the protein portion alone, catalytically inactive until it binds its required cofactor/coenzyme to form the active holoenzyme.

102. Which coenzyme derived from Vitamin B6 (Pyridoxine) is required for amino acid transamination reactions?

A. Thiamine pyrophosphate (TPP) **B. Pyridoxal phosphate (PLP)** ✓ C. Flavin adenine dinucleotide (FAD) D. Nicotinamide adenine dinucleotide (NAD⁺)

Answer: B — Pyridoxal phosphate (PLP)

Pyridoxal phosphate (from vitamin B6) is the coenzyme for transamination reactions carried out by aminotransferases.

Biochemistry: Proteins & Lipids

103. A peptide bond is chemically defined as an:

A. Ester linkage **B. Amide linkage** ✓ C. Ether linkage D. Glycosidic bond

Answer: B — Amide linkage

The peptide bond linking amino acids is chemically an amide (–CO–NH–) linkage.

104. The linear sequence of amino acids in a polypeptide chain is called the:

A. Primary structure ✓ B. Secondary structure C. Tertiary structure D. Quaternary structure

Answer: A — Primary structure

The primary structure of a protein is simply its linear sequence of amino acids.

105. Which characteristic describes the peptide bond?

A. It is easily broken by heating **B. It has partial double-bond character and is planar** ✓ C. It is highly flexible and rotates 360 degrees D. It is ionized at physiologic pH

Answer: B — It has partial double-bond character and is planar

Partial double-bond character from resonance makes the peptide bond planar and rigid, restricting rotation around it (unlike the flanking bonds).

106. Which protein is responsible for transporting oxygen in the blood?

A. Myoglobin **B. Hemoglobin** ✓ C. Transferrin D. Ferritin

Answer: B — Hemoglobin

Hemoglobin, inside red blood cells, is the main oxygen-carrying protein of the blood; myoglobin stores oxygen in muscle.

107. All are true except:

A. Fatty acid esterified with high molecular weight alcohol is wax B. Higher alcohol present in wax is cetyl C. Gangliosides are complex glycosphingolipids in brain **D. None** ✓

Answer: D — None

Waxes are indeed esters of fatty acids with long-chain alcohols such as cetyl alcohol, and gangliosides are brain glycosphingolipids — all three statements are true, so "None" (no exception) is correct.

108. Monosaccharides change from open chain to cyclic via:

A. Hydroxylation B. Carbation **C. Nucleophilic addition** ✓ D. Hydrogenation

Answer: C — Nucleophilic addition

Ring closure of an open-chain sugar occurs via intramolecular nucleophilic addition of a hydroxyl group to the carbonyl carbon, forming a hemiacetal.

109. Phospholipid in inner mitochondrial membrane (~20% of lipids):

A. Cephalin B. Plasmalogen C. Lecithin **D. Cardiolipin** ✓

Answer: D — Cardiolipin

Cardiolipin is the diphosphatidylglycerol lipid that makes up roughly a fifth of inner mitochondrial membrane phospholipids, important for oxidative phosphorylation.

110. GAG that does not form proteoglycan:

A. Heparin B. Keratan sulfate **C. Hyaluronic acid** ✓ D. Chondroitin sulfate

Answer: C — Hyaluronic acid

Hyaluronic acid is the one glycosaminoglycan that exists free in tissue and is not covalently attached to a core protein as a proteoglycan.

111. Cephaline consists of:

A. Glycerol, fatty acid, phosphoric acid, choline **B. Glycerol, fatty acid, phosphoric acid, ethanolamine** ✓
C. Glycerol, fatty acid, phosphoric acid, inositol D. Glycerol, fatty acid, phosphoric acid, serine

Answer: B — Glycerol, fatty acid, phosphoric acid, ethanolamine

Cephalin (phosphatidylethanolamine) consists of glycerol, two fatty acids, phosphoric acid and ethanolamine.

112. False statement:

A. Major fat in mammal adipose tissue is triglycerides B. Gangliosides derived from glucosylceramide contain sialic acid C. Glycosphingolipids = ceramide + sugar residues **D. Glycosphingolipids = sphingosine + galactose** ✓

Answer: D — Glycosphingolipids = sphingosine + galactose

Glycosphingolipids are built from ceramide (sphingosine + fatty acid) plus attached sugar residues, not sphingosine plus galactose alone, so option D is the false statement.

113. Incorrect about competitive inhibition:

A. Competes with substrate for active site B. Increasing substrate reverses reaction **C. No change in K_m** ✓
D. No change in V_{max}

Answer: C — No change in K_m

In competitive inhibition K_m rises (apparent affinity falls) while V_{max} is unchanged if enough substrate is added, so "no change in K_m " is the false statement.

114. Amino acid used to synthesize many hormones:

A. Valine **B. Phenylalanine** ✓ C. Alanine D. Histidine

Answer: B — Phenylalanine

Phenylalanine (via tyrosine) is the precursor for catecholamines and thyroid hormones, making it central to hormone synthesis.

115. Correct order of organelles for export:

A. RER → SER → Golgi ✓ B. SER → RER → Golgi C. RER → Golgi → SER D. Golgi → RER → SER

Answer: A — RER → SER → Golgi

Secretory proteins move RER → SER (vesicular transport) → Golgi apparatus for processing, sorting and packaging.

116. Enzyme using hydrogen acceptor as substrate:

A. Xanthine oxidase B. Aldehyde oxidase **C. Catalase** ✓ D. Tryptophan oxygenase

Answer: C — Catalase

Catalase uses hydrogen peroxide as both an electron donor and acceptor (a true oxidoreductase acting on H_2O_2 itself), matching the "hydrogen acceptor" description among the options.

117. Repeating units of hyaluronic acid are:

A. N-acetyl glucosamine and D-glucuronic acid ✓ B. N-acetyl galactosamine and D-glucuronic acid C. N-acetyl glucosamine and galactose D. N-acetyl galactosamine and L-iduronic acid

Answer: A — N-acetyl glucosamine and D-glucuronic acid

Hyaluronic acid's repeating disaccharide unit is N-acetylglucosamine plus D-glucuronic acid.

118. Phospholipid is produced by:

A. Nucleus **B. Smooth endoplasmic reticulum** ✓ C. Endoplasmic reticulum D. Cytoplasm

Answer: B — Smooth endoplasmic reticulum

Phospholipids are synthesized mainly on the smooth endoplasmic reticulum.

119. Heteropolysaccharide chondroitin differs from hyaluronic acid by presence of:

A. N-acetyl glucosamine **B. N-acetyl galactosamine** ✓ C. D-glucuronic acid D. Galactose

Answer: B — N-acetyl galactosamine

Chondroitin sulfate differs from hyaluronic acid by containing N-acetylgalactosamine instead of N-acetylglucosamine.

120. Denaturation of proteins results in:

A. Disruption of primary structure B. Breakdown of peptide bonds **C. Destruction of hydrogen bonds** ✓ D. Irreversible changes in the molecule

Answer: C — Destruction of hydrogen bonds

Denaturation disrupts secondary/tertiary/quaternary structure (weak bonds such as hydrogen bonds, ionic and hydrophobic interactions) while the primary sequence (peptide bonds) stays intact.

121. Role of cholesterol in cell membrane:

A. Acts regulating its fluidity ✓ B. Prevents formation of bumps in lipid bilayer C. Protects its integrity D. Participates in co-transport systems

Answer: A — Acts regulating its fluidity

Cholesterol's chief membrane role is regulating fluidity, keeping it within a functional range across temperatures.

122. Tertiary structure of protein describes:

A. Order of amino acids B. Location of disulfide bonds C. Loop region of protein **D. Ways of protein folding** ✓

Answer: D — Ways of protein folding

Tertiary structure is the overall three-dimensional folding pattern of a single polypeptide chain.



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